Lecture 3. Conditional Probability and Independence

Multiplication rule, total probability, Bayes's formula, independence, gambler's ruin

Contents

3.1 Conditional probability **2**

3.2 Total probability and Bayes's formula **2**

3.3 Independent events **4**

3.4 $P(\cdot \mid F)$ is a probability **5**

3.5 Selected exercises **5**

3.1 Conditional probability

Motivation. New information changes probabilities. Conditional probability restricts the sample space to what we know has occurred, and the multiplication rule lets us build complex probabilities step by step.

Definition (Conditional probability)

If $P(F) > 0$, the conditional probability of E given F is $P(E \mid F) = P(EF)/P(F)$.

With equally likely outcomes, conditioning on F makes the outcomes in F equally likely, so $P(E \mid F) = |EF|/|F|$: we simply shrink the sample space to F.

Proposition 3.1 (Multiplication rule)

$$P(E_1E_2\cdots E_n) = P(E_1)P(E_2 \mid E_1)P(E_3 \mid E_1E_2)\cdots P(E_n \mid E_1\cdots E_{n-1}).$$

Proof. Write each factor as a ratio; the product telescopes. ∎

Example 3.1

(a) An urn holds 8 red and 4 white balls; two are drawn without replacement. $P(\text{both red}) = \frac{8}{12}\cdot\frac{7}{11} = \frac{14}{33}$, which agrees with $\binom{8}{2}/\binom{12}{2}$.
(b) A deck is divided at random into four piles of 13. Let E_i be the event that the i-th pile has exactly one ace (and put the aces in piles in order). Placing the aces one at a time among the remaining slots,

$$P(E_1E_2E_3E_4) = 1\cdot\frac{39}{51}\cdot\frac{26}{50}\cdot\frac{13}{49} \approx 0.105,$$

since given that the first ace is in some pile, the second ace must be in one of the 39 of the remaining 51 positions that lie in other piles, and so on.

Example 3.2 (Matching, recursively)

In the matching problem (Example 2.4), let P_N be the probability of no match among N people and M the event that person 1 takes their own hat. Then $P_N = P(\text{no match} \mid M^c)\frac{N-1}{N}$. Given M^c, say person 1 took hat $j \neq 1$. The other $N-1$ people choose among $N-1$ hats that include hat 1 but not hat j. Either person j takes hat 1 (probability $\frac{1}{N-1}$) and the remaining $N-2$ must have no match, or we may regard hat 1 as person j's "own" hat, and then we need no match among $N-1$ people. So

$$P_N = \frac{N-1}{N}\Big(P_{N-1} + \frac{P_{N-2}}{N-1}\Big), \qquad \text{i.e.} \qquad P_N - P_{N-1} = -\frac{1}{N}(P_{N-1} - P_{N-2}).$$

With $P_1 = 0$, $P_2 = \frac{1}{2}$, induction gives $P_N - P_{N-1} = (-1)^N/N!$ and hence $P_N = \sum_{r=0}^{N}(-1)^r/r!$ once more.

3.2 Total probability and Bayes's formula

Motivation. Often we know $P(B \mid A_i)$ but want $P(A_i \mid B)$, for instance the probability of a disease given a positive test. Conditioning on the cases A_i and inverting gives Bayes's formula, whose answers are frequently counterintuitive.

Since $E = EF \cup EF^c$,

$$P(E) = P(E \mid F)P(F) + P(E \mid F^c)(1 - P(F)),$$

a weighted average of the conditional probabilities. More generally:

Theorem 3.2 (Total probability and Bayes's formula)

If $F_1, \dots, F_n$ are mutually exclusive with union S, then

$$P(E) = \sum_{i=1}^{n} P(E \mid F_i)P(F_i), \qquad P(F_j \mid E) = \frac{P(E \mid F_j)P(F_j)}{\sum_i P(E \mid F_i)P(F_i)}.$$

The $P(F_j)$ are **prior** probabilities of the hypotheses, and $P(F_j \mid E)$ are **posterior** probabilities after the evidence E. In odds form, for a hypothesis H,

$$\frac{P(H \mid E)}{P(H^c \mid E)} = \frac{P(H)}{P(H^c)} \cdot \frac{P(E \mid H)}{P(E \mid H^c)} :$$

new evidence multiplies the odds by the **likelihood ratio**.

Example 3.3 (A screening test)

A test detects a disease with probability 0.95 when it is present, but gives a false positive for 1% of healthy people. If 0.5% of the population has the disease,

$$P(D \mid +) = \frac{0.95 \cdot 0.005}{0.95 \cdot 0.005 + 0.01 \cdot 0.995} \approx 0.323.$$

Only a third of positives are diseased: the healthy are so numerous that their 1% of false positives outnumbers the true positives.

Example 3.4 (Multiple choice)

A student knows the answer with probability p and otherwise guesses among m alternatives. Then

$$P(\text{knew} \mid \text{correct}) = \frac{p}{p + (1-p)/m} = \frac{mp}{1 + (m-1)p}.$$

With $m = 5$, $p = 1/2$ this is $5/6$.

Example 3.5 (Three cards)

One card is red on both sides, one black on both sides, one red on one side and black on the other. A card is drawn at random and laid down; the upper side is red. The tempting answer $1/2$ for "the other side is red" is wrong: there are three equally likely red faces that could be on top, and two of them belong to the red–red card. Formally $P(RR \mid R \text{ up}) = \frac{1/3}{1/3 \cdot 1 + 1/3 \cdot 1/2} = \frac{2}{3}$.

Example 3.6 (Searching for a plane)

A plane is equally likely to be in one of three regions. If it is in region i, a search there misses it with probability β_i. After an unsuccessful search of region 1,

$$P(R_1 \mid \text{miss}) = \frac{\beta_1/3}{\beta_1/3 + 1/3 + 1/3} = \frac{\beta_1}{\beta_1 + 2}, \qquad P(R_j \mid \text{miss}) = \frac{1}{\beta_1 + 2} \quad (j = 2, 3).$$

The probability of region 1 decreases and the others increase, as intuition says.

For a fixed F with $P(F) > 0$, $Q(E) = P(E \mid F)$ is itself a probability, so all earlier results apply to it; e.g. $P(E_1 \mid F) = P(E_1 \mid E_2F)P(E_2 \mid F) + P(E_1 \mid E_2^cF)P(E_2^c \mid F)$.

3.3 Independent events

Definition (Independence)

E and F are **independent** if $P(EF) = P(E)P(F)$. Events $E_1, \dots, E_n$ are (mutually) independent if $P(E_{i_1} \cdots E_{i_r}) = P(E_{i_1}) \cdots P(E_{i_r})$ for every subcollection.

If E and F are independent then so are E and F^c, since $P(EF^c) = P(E) - P(E)P(F) = P(E)P(F^c)$.

Example 3.7 (Pairwise does not imply mutual)

Roll two fair dice. Let E: the sum is 7; F: the first die is 4; G: the second die is 3. Each pair is independent ($P(EF) = 1/36 = P(E)P(F)$ etc.), but $P(E \mid FG) = 1 \neq P(E)$. Note also that F_6 (sum 6) is *not* independent of F while E is: $P(\text{sum} = 7 \mid \text{first} = k) = 1/6$ for every k.

Example 3.8 (Reliability)

A parallel system works if at least one of n independent components works; component i works with probability p_i. Then $P(\text{works}) = 1 - \prod_i (1 - p_i)$. A series system works with probability $\prod_i p_i$.

Example 3.9 (E before F)

Independent trials are repeated; on each, E occurs with probability $P(E)$ and F with $P(F)$, $EF = \emptyset$. Conditioning on the first trial, $P(E \text{ first}) = P(E) + (1 - P(E) - P(F))P(E \text{ first})$, so

$$P(E \text{ occurs before } F) = \frac{P(E)}{P(E) + P(F)}.$$

In craps, a first roll of 7 or 11 wins, 2, 3 or 12 loses, and any other sum k becomes the "point", after which one rolls until k (win) or 7 (lose). With $p_k = P(\text{sum} = k)$,

$$P(\text{win}) = p_7 + p_{11} + \sum_{k \in \{4,5,6,8,9,10\}} p_k \frac{p_k}{p_k + p_7} = \frac{244}{495} \approx 0.493.$$

Example 3.10 (The problem of the points)

Trials succeed independently with probability p. What is the probability of n successes before m failures? This happens iff there are at least n successes among the first $m + n - 1$ trials (either way the contest is decided by then), so

$$P_{n,m} = \sum_{k=n}^{m+n-1} \binom{m+n-1}{k} p^k (1-p)^{m+n-1-k}.$$

This was Pascal and Fermat's question of dividing the stakes of an interrupted game: the stakes should be split in the ratio $P_{n,m} : 1 - P_{n,m}$. For instance with $p = 1/2$, $n = 4$, $m = 7$ it is 0.828.

Example 3.11 (Gambler's ruin)

A gambler with fortune i wins or loses one unit on each independent play, with probabilities p and $q = 1-p$, until reaching N or 0. Let P_i be the probability of reaching N. Conditioning on the first play, $P_i = pP_{i+1} + qP_{i-1}$, i.e. $P_{i+1} - P_i = \frac{q}{p}(P_i - P_{i-1})$, with $P_0 = 0$. Hence $P_{i+1} - P_i = (q/p)^i P_1$, and summing,

$$P_i = \begin{cases} \dfrac{1-(q/p)^i}{1-(q/p)^N}, & p \neq \frac{1}{2}, \\ \dfrac{i}{N}, & p = \frac{1}{2}, \end{cases}$$

after using $P_N = 1$ to find P_1. Letting $N \to \infty$: if $p \leq \frac{1}{2}$ the gambler is ruined with probability 1 against an infinitely rich opponent; if $p > \frac{1}{2}$ survival has probability $1 - (q/p)^i$.

3.4 $P(\cdot \mid F)$ is a probability

Since conditional probabilities obey the axioms, one may condition sequentially, and conditional independence makes sense: E_1, E_2 are conditionally independent given F if $P(E_1E_2 \mid F) = P(E_1 \mid F)P(E_2 \mid F)$.

Example 3.12 (Laplace's rule of succession)

A box contains $k+1$ coins; coin i shows heads with probability i/k, $i = 0, \ldots, k$. A coin is chosen at random and flipped n times, giving n heads. Given the coin, flips are independent, so

$$P(\text{head on flip } n+1 \mid n \text{ heads}) = \frac{\sum_i (i/k)^{n+1}/(k+1)}{\sum_i (i/k)^n/(k+1)} \approx \frac{\int_0^1 x^{n+1}dx}{\int_0^1 x^n dx} = \frac{n+1}{n+2}$$

for large k. The flips are not independent unconditionally: early heads are evidence that the coin is good.

Example 3.13 (Updating sequentially)

Hypotheses $H_1, \ldots, H_n$ have priors $P(H_i)$, and pieces of evidence $E_1, \ldots, E_k$ are conditionally independent given each H_i. Then

$$P(H_i \mid E_1 \cdots E_k) \propto P(H_i) \prod_{j=1}^{k} P(E_j \mid H_i),$$

so the posterior after E_1 serves as the prior for E_2, and so on; the order in which the evidence arrives does not matter.

3.5 Selected exercises

Motivation. These are Ross's theoretical exercises for Chapter 3 (book numbering). Each one is solved by the main idea of the lecture: condition on a well-chosen first or last event and obtain a recursion or a symmetry.

Exercise 3.7(a) (The colour that survives)

An urn contains $n \geq 1$ white and $m \geq 1$ black balls. The balls are withdrawn one at a time, at random, until only balls of one colour are left. Show that they are all white with probability $n/(n+m)$.

Solution.

Step 1. Imagine that the withdrawals continue until the urn is empty, and label the balls. All $(n+m)!$ orders of withdrawal are equally likely, and the actual experiment is the initial part of this order up to the stopping time.

Step 2. When we stop, the balls left in the urn are those in a final segment of the order. This segment is nonempty, because at the start both colours are present and so we never stop before at least one ball remains. Hence it contains the last ball of the order, and all the balls left have the colour of that last ball. So the balls left are white iff the last ball of the full order is white.

Step 3. For each fixed ball, exactly $(n+m-1)!$ orders end with it, so each ball is last with probability $1/(n+m)$. With n white balls, $P(\text{last ball white}) = n/(n+m)$. ∎

Exercise 3.16 (An even number of successes)

Let P_n be the probability that n Bernoulli trials with success probability p result in an even number of successes (0 counts as even). Show that

$$P_n = p(1 - P_{n-1}) + (1-p)P_{n-1} \quad (n \geq 1),$$

and deduce $P_n = (1 + (1-2p)^n)/2$.

Solution.

Step 1. Let S be the number of successes in the first $n-1$ trials. The total is even iff either trial n succeeds and S is odd, or trial n fails and S is even. These two events are disjoint, and trial n is independent of the first $n-1$ trials, so

$$P_n = p\,P(S \text{ odd}) + (1-p)P(S \text{ even}) = p(1 - P_{n-1}) + (1-p)P_{n-1}.$$

Step 2. The recursion reads $P_n = p + (1-2p)P_{n-1}$. For $n = 0$ there are no successes, so $P_0 = 1 = (1 + (1-2p)^0)/2$. If $P_{n-1} = (1 + (1-2p)^{n-1})/2$, then

$$P_n = p + \frac{1-2p}{2} + \frac{(1-2p)^n}{2} = \frac{1 + (1-2p)^n}{2},$$

which completes the induction. ∎

Exercise 3.18 (No run of three heads)

Let Q_n be the probability that no run of three consecutive heads appears in n tosses of a fair coin. Show that

$$Q_n = \tfrac{1}{2}Q_{n-1} + \tfrac{1}{4}Q_{n-2} + \tfrac{1}{8}Q_{n-3} \quad (n \geq 3), \qquad Q_0 = Q_1 = Q_2 = 1,$$

and find Q_8.

Solution.

Step 1. Let R be the event of no run of three heads and F_j $(j = 1, 2, 3)$ the event that the first tail occurs at toss j; let H be the event that tosses $1, 2, 3$ are all heads. Then F_1, F_2, F_3, H partition the sample space and $P(F_j) = 2^{-j}$.

Step 2. On H there is a run, so $RH = \emptyset$. For $j \le 3$, on F_j the first j tosses are $j-1$ heads followed by a tail and contain no run; a run of three heads cannot straddle toss j, which is a tail. Hence $RF_j = F_j \cap R_j$, where R_j is the event of no run of three heads among tosses $j+1, \dots, n$.

Step 3. R_j depends only on tosses $j+1, \dots, n$, so it is independent of F_j, and these $n-j$ tosses are again independent fair tosses, so $P(R_j) = Q_{n-j}$. Therefore

$$Q_n = \sum_{j=1}^{3} P(F_j)P(R_j) = \tfrac{1}{2}Q_{n-1} + \tfrac{1}{4}Q_{n-2} + \tfrac{1}{8}Q_{n-3}.$$

Step 4. Iterating, $Q_3 = \frac{7}{8}$, $Q_4 = \frac{13}{16}$, $Q_5 = \frac{24}{32}$, $Q_6 = \frac{44}{64}$, $Q_7 = \frac{81}{128}$ and

$$Q_8 = \tfrac{1}{2}\cdot\tfrac{81}{128} + \tfrac{1}{4}\cdot\tfrac{44}{64} + \tfrac{1}{8}\cdot\tfrac{24}{32} = \frac{81+44+24}{256} = \frac{149}{256} \approx 0.582.$$

■

Exercise 3.21 (The ballot problem)

In an election, candidate A receives n votes and candidate B receives m votes, $n > m \ge 0$, and all $\binom{n+m}{n}$ orderings of the votes are equally likely. Let $P_{n,m}$ be the probability that A is strictly ahead throughout the count.
(a) Derive a recursion for $P_{n,m}$ in terms of $P_{n-1,m}$ and $P_{n,m-1}$ by conditioning on who receives the last vote.
(b) Prove that $P_{n,m} = (n-m)/(n+m)$.

Solution.

(a)

Step 1. Each ordering of n A's and m B's that ends with A corresponds to exactly one ordering of $n-1$ A's and m B's (delete the last letter), so the last vote is A's with probability $\binom{n+m-1}{n-1}/\binom{n+m}{n} = n/(n+m)$, and given this the first $n+m-1$ votes are a uniformly random ordering of $n-1$ A's and m B's. The same holds with the roles exchanged when the last vote is B's.

Step 2. Since $n > m$, A is ahead after the final vote, so A is ahead throughout iff A is ahead throughout the first $n+m-1$ votes. Hence for $m \ge 1$

$$P_{n,m} = \frac{n}{n+m}P_{n-1,m} + \frac{m}{n+m}P_{n,m-1},$$

with the conventions $P_{k,k} = 0$ for $k \ge 1$ (a count ending in a tie cannot keep A strictly ahead) and $P_{n,0} = 1$.

(b)

Step 1. We use induction on $n+m$. If $m = 0$, every vote is A's and $P_{n,0} = 1 = n/n$. The formula also gives 0 when $n = m$, matching the convention for $P_{k,k}$.

Step 2. Let $n > m \ge 1$ and assume the formula for all pairs with smaller total (it covers $P_{n-1,m}$ also when $n-1 = m$). By (a),

$$\begin{aligned} P_{n,m} &= \frac{n}{n+m}\cdot\frac{n-1-m}{n+m-1} + \frac{m}{n+m}\cdot\frac{n-m+1}{n+m-1} \\ &= \frac{(n-m)(n+m) - (n-m)}{(n+m)(n+m-1)} = \frac{n-m}{n+m}, \end{aligned}$$

where we used $n(n-m-1) + m(n-m+1) = n^2 - m^2 - (n-m)$. ■